An IoT-Enabled System for Real-Time Fuel Monitoring, GPS Tracking, and Anomaly Detection in Trucks Using Non-Parametric Unsupervised Machine Learning

---

A Thesis Proposal
Presented to the Faculty of the
Department of Electronics and Computer Engineering
Gokongwei College of Engineering
De La Salle University

In Partial Fulfillment of the
Requirements for the Degree of
Bachelor of Science in Computer Engineering

---

by

BARJA Samuelle P.
CASTELO Konrad Romel C.
DE LA ROSA Eliza Rica R.
MONTEMAYOR Alyssa S.

May, 2026

ABSTRACT

The logistics and transportation industry continues to face challenges in fuel management, anomaly detection, and real-time fleet monitoring. Existing solutions often operate in silos, resulting in fragmented data, undetected fuel irregularities, and delayed maintenance responses. This study proposes an IoT-based, context-aware system designed to monitor real-time fuel levels, track GPS location, and detect anomalous fuel consumption in trucks using non-parametric unsupervised machine learning models, including Matrix Profile and Isolation Forest. The system leverages CAN Bus/OBD-II interfacing for ECU fuel data, GPS modules for location logging, LoRa and GSM modules for reliable IoT-based communication, and embedded alert hardware for driver rest and maintenance notifications. Sensor and vehicle data are synchronized and transmitted to a cloud-based dashboard where anomalies are automatically flagged, and actionable alerts are generated. The system will be evaluated in real-world driving scenarios to measure detection accuracy, false positive rates, and response latency. This integrated platform aims to enhance operational safety, improve fuel efficiency, and enable proactive, data-driven fleet management decisions.

Index Terms—Fuel Monitoring, GPS Tracking, Anomaly Detection, Internet Of Things, Unsupervised Model Learning, Isolation Forest, Matrix Profile

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

ABBREVIATIONS

NOTATION

FN	false negative	70
FP	false positive	70
L	data transmission latency	71
T_{actual}	timestamp of actual anomaly occurrence	71
T_{detected}	timestamp when an anomaly is detected	71
TN	true negative	70
TP	true positive	70

141 GLOSSARY

142	cloud analytics	Data analysis performed on cloud-based platforms, enabling real-time access to fleet data (e.g., GPS and fuel metrics) and supporting informed decision-making.
143	deep learning	A subset of machine learning involving deep neural networks with multiple layers. Used in predictive analytics and anomaly detection in real-time fleet systems.
144	fleet management	The administration and coordination of commercial vehicle operations, including monitoring of fuel use, driver behavior, vehicle maintenance, and routing.
145	Isolation Forest	An unsupervised machine learning algorithm for anomaly detection that isolates outliers based on random partitioning of data points.
146	Matrix Profile	A time-series data mining method used for pattern recognition and anomaly detection in sequential datasets like fuel usage over time.
147	unsupervised machine learning	A type of machine learning where the model learns patterns from unlabeled data. Useful for anomaly detection where normal vs. abnormal behavior is not predefined.

Chapter 1

INTRODUCTION

1.1 Background of the Study

The transportation and logistics industry is a critical driver of economic development but is increasingly facing operational inefficiencies and security challenges, especially in fuel management and vehicle tracking. Fuel theft, inaccurate fuel monitoring, inefficient routing, and lack of real-time data remain persistent issues for fleet operators, particularly in developing countries. These challenges are exacerbated by the absence of scalable, integrated technological systems that provide reliable data-driven solutions [Ahmed et al., 2017, Alquhali et al., 2019, Crisgar, 2021].

Fuel is one of the most substantial and recurring expenses in logistics operations, with its accurate monitoring being essential to maintaining profitability and operational efficiency. Even minor inconsistencies in fuel tracking—such as errors in manual readings, unreported consumption, or unnoticed leakage—can accumulate into significant losses over time. Komal Shoukat Ali et al. (2018) [Komal Shoukat Ali, 2018] report that fuel-related fraud and inefficiencies can contribute up to 30% of the total operational overhead in large fleet operations. These losses are often compounded by issues such as unauthorized vehicle use, fuel siphoning, and unoptimized driving routes, which not only drive up fuel costs but also disrupt delivery schedules, vehicle maintenance cycles, and driver accountability. Despite efforts to manage these concerns, many logistics providers continue to rely on fragmented or outdated systems that offer limited visibility into fuel-related metrics and driver behavior [Dhivyasri and Mariappan, 2015, Obikoya, 2014].

In addition to financial implications, traditional fuel monitoring systems struggle to detect abnormal fuel consumption patterns in real time. These anomalies may be caused by aggressive or inefficient driving habits, extended engine idling, mechanical malfunctions (such as injector faults or fuel line leaks), or variable environmental conditions like terrain and traffic. However, without intelligent tools that correlate fuel data with contextual factors—such as vehicle speed, route topography, and time of day—fleet managers are often unable to distinguish between acceptable fuel use and problematic deviations. As highlighted by Mumcuoglu et al. (2023) [Mumcuoglu, 2023], conventional systems lack the sophistication to perform this level of analysis, leading to delayed detection of inefficiencies and reduced responsiveness. This underscores the urgent need for smart, data-driven solutions that combine fuel monitoring with GPS tracking and anomaly detection models to proactively manage consumption and enhance overall fleet performance [Barbado and Corcho, 2022, Beteto, 2022, Fortela, 2023].

To address these problems, researchers and engineers are increasingly exploring the convergence of IoT, GPS, and AI—particularly machine learning—to develop smart, real-time monitoring systems [Alquhali et al., 2019, Joshi, 2024, Prabhu et al., 2022]. These systems aim to optimize fuel usage, enhance fleet security, and improve operational efficiency through automation and predictive analytics.

The transportation and logistics sector has attracted significant scholarly attention, particularly in the development of smart systems for operational efficiency and security. The academic literature reflects a growing interest in leveraging IoT, GPS, and AI technologies to enhance fleet management, particularly in areas such as fuel monitoring, vehicle tracking, and anomaly detection. Several studies in recent years have investigated IoT-enabled fuel monitoring, GPS tracking, and anomaly detection solutions [G, 2022, G, 2023, Patil and Patil, 2024, Prabhu et al., 2022, RS et al., 2024, S et al., 2024].

- **Intelligent IoT-Enabled Real-Time Monitoring System for Logistics Management:** Joshi et al. [Joshi, 2024] developed a hybrid logistics management system that integrates IoT sensors, GPS, Radio Frequency Identification (RFID), and cloud analytics. Their solution achieved real-time tracking of assets and vehicles, enhancing transparency and decision-making in logistics operations. However, the study was limited by the type of hardware used and did not include advanced AI components for anomaly detection.

- **Fuel Consumption Classification for Heavy-Duty Vehicles:** Mumcuoglu et al. [Mumcuoglu, 2023] employed machine learning to classify fuel consumption anomalies based on naturalistic driving data from 57 trucks. With an accuracy of 92.2%, their approach effectively identified abnormal fuel consumption linked to driver behavior. Nevertheless, the dataset was relatively small (606 records), and integration with real-time vehicle diagnostics was not tested.

- **Development of an Intelligent Fuel Monitoring and Theft Detection System Equipped with GPS & GSM Integration:** Mathi et al. [Mathi, 2024] introduced an Internet of Things (IoT)-based solution combining resistance-based fuel level sensors, GPS, Global Systems for Mobile Communications (GSM), and real-time alarm mechanisms. Their system successfully monitored fuel and battery anomalies in hybrid vehicles but lacked higher-precision sensors and deep learning integration for smarter detection.

Despite these advancements, several limitations and research gaps remain:

- **Scalability Constraints:** Many existing solutions are limited to small-scale deployments or specific vehicle types (e.g., motorcycles, generators) and have not been validated for large truck fleets in diverse environments [RS et al., 2024, Tatababu et al., 2024].

- **Data and Sensor Limitations:** Several studies rely on basic sensors (resistance, ultrasonic) that may lack the accuracy and robustness needed for high-volume commercial operations. Furthermore, datasets used are often small or simulated, limiting generalizability [Nada, 2017, aut, 2019].

- **Fragmented Solutions:** Many systems only address a single issue — fuel monitoring, GPS tracking, or theft detection — without offering a comprehensive, unified solution integrating these components using AI and IoT [G, 2022, Crisgar, 2021].

These gaps underline the need for a robust, scalable, and integrated IoT-enabled system capable of monitoring fuel levels in real time, tracking vehicle location, and detecting operational anomalies using machine learning algorithms. Such a system holds the potential to significantly reduce fuel fraud, enhance operational efficiency, and support more informed decision-making in fleet management. This study focuses on designing and implementing such a system within an active fleet operation, specifically addressing the technical and logistical challenges encountered by commercial truck fleets operating across varied conditions. By leveraging IoT sensors, GPS modules, and unsupervised machine learning-based analytics, the proposed solution aims to deliver a comprehensive, automated, and intelligent monitoring platform that advances the current landscape of smart logistics technologies.

1.2 Prior Studies

TABLE 1.1 LIST OF PRIOR STUDIES

Literature Title (APA Citation)	Research Problem	Method	Results	Limitations
Intelligent IoT-Enabled Real-Time Monitoring System for Logistics Management [Joshi, 2024]	Logistics systems lack integration and real-time monitoring capabilities for secure and efficient operations.	Hybrid IoT-based logistics management combining sensors, GPS, RFID, and cloud analytics for real-time monitoring and decision-making.	Provides real-time tracking of assets, vehicles, and equipment.	Constrained by hardware type and limited test cases; lacks advanced AI for anomaly detection.
Real Time Automatic Vehicle Monitoring System Using IoT [G, 2022]	Lack of real-time vehicle tracking and passenger/driver identity verification in institutional transport.	Used RFID, GPS, and NodeMCU ESP8266 with cloud storage and mobile app.	Enabled real-time tracking of location, speed, and passenger count.	Limited to institutional use; lacks anomaly detection and commercial scalability.
GPS-Based Vehicle Tracking and Theft Detection Systems using Google Cloud IoT Core & Firebase [Crisgar, 2021]	Ineffective vehicle recovery due to delayed or limited GPS/GSM tracking.	Implemented GPS-based tracking using Google Cloud IoT Core, MQTT, Firebase, and PWA interface.	Enabled real-time vehicle tracking, theft detection, and remote monitoring.	Designed for passenger cars; may require hardware adaptation for trucks/fleets.
Fuel Quantity and Quality Detection in Automobiles Using IoT [RS et al., 2024]	Fuel fraud through inaccurate dispensing and poor fuel quality in motorcycles.	IoT-enabled fuel management using strain gauges and quality sensors with real-time display/logging.	Achieved accurate real-time quantity and quality measurement.	Designed for motorcycles; not validated for larger fleets.
Fuel Consumption Classification for Heavy-Duty Vehicles [Mumcuoglu, 2023]	Identifying abnormal fuel consumption due to driver behavior or system faults.	Machine learning models using driving data from 57 trucks, considering weight and road slope.	92.2% accuracy; F1 score 0.78 for outlier detection.	Dataset limited to 606 records; lacks real-time integration.

Literature Title (APA Citation)	Research Problem	Method	Results	Limitations
Automatic Fuel Tank Monitoring, Tracking & Theft Detection System [Komal Shoukat Ali, 2018]	Inefficiency of manual fuel monitoring.	Fuel sensors, Arduino, GSM, GPS, LabView for data logging.	Real-time monitoring, alerts for theft/leakage.	Tested on basic hardware; lacks advanced analytics and large-scale validation.
Development of an Intelligent Fuel Monitoring and Theft Detection System [Mathi, 2024]	Fuel theft and battery anomalies in hybrid/electric vehicles.	Resistance-based fuel sensor, voltage sensor, GPS, GSM, buzzer, LCD.	Reliable monitoring and anomaly detection.	Lacks advanced sensors for enhanced leak detection.
Unsupervised Machine Learning to Detect Impending Anomalies in Testing of Fuel Economy and Emissions [Fortela, 2023]	Impending anomalies in fuel economy and emissions testing of light-duty vehicles.	Unsupervised ML methods (PCA, K-Means, SOM) on 34,602 test samples.	Trend detection in large datasets.	Limited to historical datasets; not validated on real-time operational data.
Smart Fuel Monitoring System for Diesel Generator [Tatababu et al., 2024]	Fuel quantity indication and theft in generators.	Ultrasonic/capacitive sensors, flow sensors, GSM, GPS, intelligent theft detection.	Accurate real-time monitoring, theft prevention.	Findings specific to tested generators; limited generalizability.
Design and Implementation for Trucks Tracking System Using GPS Based on Semantic Web [Nada, 2017]	Inefficient GPS-based truck tracking for fleet management.	GPS tracking system integrated with semantic web platform.	Accurate location tracking and reporting via web interface.	Dependent on GSM/GPS reliability; lacks IoT integration and anomaly detection.

Literature Title (APA Citation)	Research Problem	Method	Results	Limitations
Automatic Fuel Monitoring System [aut, 2019]	Accurate fuel monitoring and theft prevention.	Fuel sensors and GSM module with SMS alerts.	Monitors fuel, triggers alarms, and sends GPS alerts.	Basic sensors; limited scalability and analytics in complex environments.
Interpretable Machine Learning Models for Vehicle Fuel Consumption Anomalies [Barbado and Corcho, 2022]	Detecting/explaining fuel anomalies in fleets to optimize usage.	Unsupervised anomaly detection combined with interpretable ML.	Up to 80% anomalous instances explained; potential fuel reduction up to 88%.	Only analyzed individual feature relevance; not fully integrated in real-time systems.

1.2.1 Research Gaps Identified

A review of existing literature in fuel monitoring and anomaly detection for logistics and fleet management reveals persistent gaps that point to a need for more robust, integrated solutions. Although many systems incorporate IoT elements—such as sensors, GSM, and GPS—these are often limited in scope, typically designed for specific applications like diesel generators or motorcycles, rather than being adaptable for broader fleet use.

For example, systems like the Smart Fuel Monitoring System for Diesel Generators [Tatababu et al., 2024] and the Automatic Fuel Monitoring System [aut, 2019] demonstrate basic real-time monitoring and theft detection capabilities. However, these systems heavily rely on elementary sensor technology and offer minimal integration with advanced analytics or anomaly detection methods. Similarly, the machine learning-based approach in [Mumcuoglu, 2023] was limited in its use of data and did not incorporate real-time or unsupervised learning techniques.

While [Barbado and Corcho, 2022] explored the potential of interpretable machine learning and unsupervised anomaly detection, their models were evaluated on constrained datasets and not integrated into real-time operational systems. [Fortela, 2023] also demonstrated effective anomaly detection using unsupervised learning in emissions and fuel economy testing, but again, their data was static and gathered in controlled environments. These findings were not validated using on-vehicle, operational data, which limits their applicability in dynamic, real-world conditions.

Additionally, studies such as those by [Joshi, 2024] and [Nada, 2017] underscore the need for real-time GPS tracking in fleet operations, but they stop short of presenting a cohesive system that incorporates GPS, fuel monitoring, and intelligent anomaly detection into one unified platform.

The core research gap, therefore, lies in the absence of a fully integrated, real-time, IoT-based system that combines fuel monitoring, GPS tracking, and anomaly detection powered by unsupervised machine learning. Addressing this gap is vital to improving operational efficiency, fuel security, and predictive maintenance in modern fleet management.

1.2.2 to Address the Gaps

This study proposes the development of an integrated IoT-based system that unifies fuel level monitoring, GPS-based vehicle tracking, and unsupervised machine learning for anomaly detection. Unlike previous solutions that focus on isolated functions or static datasets, this system is designed with real-time operation in mind. During initial testing, data will be simulated using a custom-built fuel tank setup to enable controlled evaluation of the system's performance and anomaly detection capabilities.

The system architecture will incorporate scalable and cost-efficient components such as , GPS modules, and cloud-based transmission via mobile networks, ensuring compatibility with a variety of fleet vehicles and usage environments.

For anomaly detection, the system will employ unsupervised, non-parametric machine learning algorithms like Matrix Profile and Isolation Forest. These models excel in identifying outliers within time-series data without the need for labeled datasets—making them ideal for fleet scenarios where data behavior varies with routes, driver habits, or environmental conditions. Unlike traditional parametric approaches, non-parametric models are more flexible and adaptive, as they do not assume any fixed data distribution.

In addition, a context-aware dashboard will be developed to provide fleet managers with meaningful insights by correlating fuel usage patterns with routes and travel times. The dashboard will support real-time notifications such as driver rest alerts, maintenance prompts, and efficiency summaries—enabling data-informed decision-making and operational improvements.

By addressing the limitations of prior research—especially the lack of integration, real-time analysis, and adaptive anomaly detection—this study offers a practical and scalable solution for enhancing fuel management and fleet operations in dynamic environments.

1.3 Problem Statement

Efficient fuel monitoring and route tracking are essential components of sustainable and cost-effective truck fleet operations. However, existing systems face significant challenges, including inaccurate fuel measurement, disconnected data sources, delayed anomaly detection, and limited support for safety-based decision-making, as outlined below.

1. PS1: The Ideal Scenario
    - Trucking companies should have real-time access to accurate fuel usage and GPS tracking data to ensure efficient operations, reduce fuel theft, and improve route planning.
    - Drivers should receive timely rest alerts and vehicle maintenance reminders based on actual driving and fuel metrics to promote safety, avoid overfatigue, and extend vehicle life.
    - Fleet managers should be able to detect unusual fuel consumption patterns automatically, enabling early intervention and optimized scheduling.
    - All of these should operate through automated, cloud-based systems that integrate IoT sensors, machine learning, and remote dashboard monitoring.
2. PS2: The Reality of the Situation
    - Most trucking operations lack real-time visibility into fuel usage, relying instead on manual readings or delayed reports that are prone to error and manipulation.
    - GPS tracking is often siloed from other performance indicators like fuel consumption and driving behavior, making behavior-based analysis or route optimization difficult.
    - Anomalous fuel consumption events (e.g., theft, leakage, or inefficiency) are rarely detected until long after they've occurred, resulting in financial loss and operational delays.

- Rest and maintenance schedules are typically based on fixed intervals rather than real-time usage data, increasing the risk of driver fatigue, breakdowns, and missed service opportunities.
- Existing technologies are either too costly, fragmented, or lack contextual intelligence to automatically correlate GPS, fuel, and operational metrics.

3. PS3: The Consequences for the Audience

- Truck fleet operators face increasing operational costs due to undetected fuel inefficiencies and theft, impacting business sustainability.
- Driver safety is compromised when rest is not scheduled based on actual driving patterns or system recommendations.
- Vehicle lifespan shortens and maintenance costs rise when alerts are not tied to real-time usage, resulting in unexpected breakdowns and downtime.
- Decision-makers are unable to make informed, data-driven decisions due to fragmented systems and the absence of an integrated monitoring and alert platform.
- These issues collectively hinder operational efficiency, safety, and environmental responsibility, particularly for logistics providers in developing regions lacking advanced fleet management tools.

1.4 Objectives and Deliverables

1.4.1 General Objective (GO)

GO: To develop and evaluate an IoT-based system for trucks, in collaboration with the partner truck company Hino Parañaque Subic GS Auto Inc. (SGSA), that monitors real-time fuel levels, tracks routes using GPS, and detects unusual fuel consumption using machine learning, with the goal of improving driver safety, supporting rest scheduling, and facilitating proactive vehicle maintenance through automated alerts using fuel and distance metrics, based on established service guidelines.;

1.4.2 Specific Objectives (SOs)

- SO1: To measure fuel levels using digital float sensors integrated through the truck's ECU/diagnostic connector, achieving ±5% accuracy compared to manual readings in controlled conditions.
- SO2: To transmit real-time fuel and GPS data through IoT-based communication modules to a cloud server and central web dashboard with latency not exceeding 3 seconds.
- SO3: To capture and log GPS location data every 5 seconds via an embedded module, correlating route information with fuel and time data for behavior tracking.
- SO4: To design and evaluate a context-aware fuel usage deviation modeling approach using non-parametric unsupervised machine learning models, like Matrix Profile or Isolation Forest, achieving 90% precision and 10% false

positives in identifying anomalous fuel consumption patterns within controlled environments.

- SO5: To develop and integrate hardware-based rest alerts and maintenance reminders for long-haul driver safety linked to a cloud-based dashboard hosted at the partner trucking company's base, using cumulative fuel and distance data to guide break schedules and service intervals (300 km or 6-hour rest recommendations, 5,000 km oil check reminders).

1.4.3 Expected Deliverables

1.5 Expected Deliverables per Objective

TABLE 1.2 EXPECTED DELIVERABLES PER OBJECTIVE

Objectives	Expected Deliverables
GO: To develop and evaluate an IoT-based system for trucks in collaboration with Hino Parañaque Subic GS Auto Inc. (SGSA) that monitors real-time fuel levels, tracks routes using GPS, and detects unusual fuel consumption using machine learning, improving driver safety, supporting rest scheduling, and facilitating proactive vehicle maintenance through automated alerts.	a) **Trained Unsupervised Machine Learning Model** • A validated ML model (Matrix Profile or Isolation Forest) trained on SGSA-reflective datasets for detecting anomalous fuel consumption. b) **Web-Based Monitoring Interface** • Real-time dashboard showing GPS location, fuel level, distance, and operating hours for each truck. c) **Embedded In-Truck HMI** • Driver-friendly interface displaying driver credentials (name, PIN, and assigned truck), current trip activity, and alert notifications. d) **Cloud Backend Infrastructure** • Cloud-based data aggregation, processing, and automated alert generation accessible from SGSA's base. e) **Performance Documentation** • Complete validation results under laboratory and SGSA field conditions.
SO1: To measure fuel levels using digital float sensors integrated through the truck's ECU/diagnostic connector, achieving ±5% accuracy compared to manual readings in controlled conditions.	a) Calibrated fuel-level acquisition module interfaced through the truck ECU/diagnostic connector. b) Accuracy validation report comparing sensor data with manual dipstick levels. c) Embedded firmware for fuel data acquisition, filtering, and preprocessing.

Objectives	Expected Deliverables
SO2: To transmit real-time fuel and GPS data through IoT-based communication modules to a cloud server and central web dashboard with latency not exceeding 10 seconds.	a) Integrated IoT communication module with GSM as the primary communication path, with local cache buffering and delayed synchronization to ensure offline data continuity. b) Cloud backend with secure API endpoints for receiving fuel and GPS data. c) Latency benchmarking report under variable network environments. d) Optimized MQTT/HTTP transmission logic with buffering and fail-safe retries. e) Security documentation covering packet integrity and encryption features.
SO3: To capture and log GPS location data every 5 seconds via an embedded module, correlating route information with fuel and time data for behavior tracking.	a) Integrated GPS module capturing coordinates and timestamps at 5-second intervals. b) Data fusion algorithm linking GPS location, fuel levels, and time progression. c) Route visualization through a map interface on the dashboard. d) Database structure storing synchronized fuel–location–time datasets. e) Behavioral analytics showing fuel usage per route segment.
SO4: Design and evaluate a context-aware fuel usage deviation modeling approach using non-parametric unsupervised ML models (Matrix Profile or Isolation Forest), achieving 90% precision and 10% false positives.	a) Labeled normal and anomalous fuel consumption datasets. b) Implementation of Isolation Forest and/or Matrix Profile in Python/MATLAB. c) Evaluation metrics report including precision, recall, and FPR 10%. d) Exportable model (PMML/ONNX) for real-time pipeline integration. e) Context-aware analytics combining slope, route, load, and time-of-day factors.

Objectives	Expected Deliverables
SO5: To develop and integrate hardware-based rest alerts and maintenance reminders for long-haul driver safety linked to a cloud-based dashboard hosted at the partner trucking company's base, using cumulative fuel and distance data to guide break schedules and service intervals (300 km or 6-hour rest recommendations, 5,000 km oil check reminders).	a) Cloud dashboard showing fuel usage, distance, operating hours, and maintenance status. b) Rule-based alert engine for 300 km / 6-hour rest reminders and 5,000 km oil maintenance alerts. c) Physical embedded module: Start/Stop button, buzzer/light indicators for rest and maintenance alerts. d) Backend logic computing distance-based and time-based safety thresholds. e) Dashboard filters for alert logs, historical events, and driver compliance summaries.

1.6 Significance of the Study

1.6.1 Technical Benefit

This research will introduce a context-aware, cloud-connected fleet monitoring system that integrates digital float sensors, GPS route tracking, and unsupervised machine learning to enhance real-time monitoring of fuel usage and detection of anomalous consumption events. Fuel levels will be measured using digital float sensors interfaced through the truck's ECU or diagnostic connector, with the system targeting an accuracy of ±5% compared to manual readings under controlled test conditions. Real-time fuel and GPS data will be transmitted to a cloud server with a maximum latency of three seconds using IoT-based communication modules, ensuring continuous data availability to fleet managers.

GPS modules embedded in the vehicles will log location data every 5 seconds, enabling the system to correlate fuel consumption with time, distance, and route conditions for behavioral tracking and operational analysis. The study will implement non-parametric unsupervised learning methods—specifically Matrix Profile and Isolation Forest—to identify deviations in fuel usage without the need for labeled datasets, allowing scalable and adaptive anomaly detection. In addition, a cloud-based dashboard will integrate hardware-driven rest alerts and maintenance reminders, leveraging cumulative fuel and distance data to generate intelligent break schedules and service interval notifications. Through these combined capabilities, the system aims to improve technical performance by enhancing measurement accuracy, communication responsiveness, anomaly detection reliability, and overall automation in fleet fuel management and driver safety support.

1.6.2 Social Impact

Through an implementation of automatic rest alerts and preemptive maintenance cues, the study encourages safety and health of the drivers to prevent accidents commonly caused by fatigue as well as breakdowns. The system will feature the ability to inform drivers when they are near critical or recommended limits (e.g. 300 km of continuous driving or 6 hours of operation) this will encourage healthy driving practices and compliance with regulations, especially for long distance hauls. These practices will empower the fleet managers who can systematically enforce them, relaying the culture of safety and responsibility into the industry.

In addition, the GPS-based route monitoring and fuel usage logging will help reinforce accountability and transparency, two essentials in preventing fuel misuse. This will help to build trust between management and drivers, and also allow the transport logistics to be data led in the decision making.

1.6.3 Environmental Welfare

The system will improve operational efficiency through minimizing waste in fuel, optimizing route performance and decreasing unplanned maintenance downtime. Through precise anomaly detection and maintenance forecasting, the platform enables cost effective operation of the fleet and longer life of vehicles. It will also help achieve environmental sustainability by minimizing unneeded fuel consumption and emissions from inefficient routes, fuel leaks or late maintenance. By promoting timely service interventions (e.g. oil changes at every 5,000 kilometers) the study aims to support cleaner engine performance and reduced environmental impact over time.

1.7 Assumptions, Scope, and Delimitations

1.7.1 Assumptions

a) Technical Infrastructure

- Consistent internet connection is required to maintain real time data exchange between the IoT sensors, GPS module, and cloud sensor.
- The system integration between the hardware and cloud based infrastructure operates smoothly without significant technical issues.

- The system will have access to a stable power supply for an uninterrupted operation of the on board IoT system.
- Mobile network coverage is able to handle a low latency data communication (within a 3 second threshold).

b) Data Accuracy

- Digitally calibrated float sensors integrated through the truck's ECU are able to provide fuel-level measurements with a validated accuracy of ±5% when compared to manual dipstick readings.
- The GPS module captures and logs precise coordinates at 5-second intervals, enabling accurate route reconstruction and correlation with fuel consumption and time data.

c) Machine Learning Model Performance

- The unsupervised learning models (e.g., Matrix Profile, Isolation Forest) are capable of identifying anomalous fuel consumption with 90% precision and 10% false positives in controlled environments.
- The training and evaluation data of anomaly detection will be representative of normal fuel consumption and truck behavior under different operating conditions.

d) User Interaction

- It is assumed that users (e.g., truck drivers, fleet operators, and system administrators) will interact with the system according to their roles and available tools.

- Truck drivers are not required to use personal mobile devices; instead, they will rely on a simple onboard interface consisting of a small display and buzzer for receiving rest alerts, maintenance reminders, and status notifications.
- Fleet operators and system administrators will interact with the cloud-based dashboard using standard office computers to monitor fuel data, GPS routes, alerts, and maintenance logs.
- Fleet personnel are expected to follow system recommendations (e.g., rest intervals, maintenance reminders) based on real-time fuel, distance, and performance metrics.

e) Operating Environment

- Environmental conditions, together with power and network breakdowns, will not significantly affect system performance.
- System maintenance is performed routinely and at normal intervals to ensure operational continuity.

1.7.2 Scope

The scope of the study encompasses the development and implementation of a smart fleet monitoring system that utilizes IoT and machine learning technologies for enhanced fuel usage , route monitoring and vehicle maintenance reminder.

a) **Real-Time Fuel Monitoring**

i. Deployment of digitally calibrated float sensors integrated through the truck's ECU, providing continuous fuel-level monitoring with a validated

±5% accuracy compared to manual dipstick measurements.

b) **Cloud Based Data Transmission**

i. Implementation of mobile network communication protocols to transmit data from the vehicle to a centralized cloud server with a latency not exceeding 3 seconds.

c) **GPS Enabled Route Tracking**

i. Incorporation of GPS modules for logging vehicle location at a sampling rate of 5 seconds, allowing correlation of fuel consumption with route patterns and driving behavior.

d) **Anomaly Detection via Machine Learning**

i. Context-aware fuel deviation detection model development by means of unsupervised learning methods (e.g., Isolation Forest, Matrix Profile).

ii. Analysis of precision and false positive rate of the model in a controlled setting.

e) **Driver Rest Alerts and Maintenance Reminders**

i. Incorporation of a dashboard system deployed at the trucking company's base to show cumulative fuel and distance measures.

ii. Production of automated rest alerts (e.g. every 300 km or after 6 hours) and maintenance reminders (e.g. after every 5,000 km for checking oil).

1.7.3 Delimitations

a) **Focus on Fuel Monitoring and Anomaly Detection**

- The system will be designed to focus specifically on fuel level monitoring, detecting consumption anomalies, and providing maintenance guidance based on fuel usage and distance traveled. It will not incorporate other real-time diagnostic data from the vehicle's onboard systems.

b) **Hardware and Communication Constraints**

- The system utilizes digitally calibrated float sensors, an embedded GPS module, and a microcontroller interfaced through the truck's ECU/diagnostic connector; higher-end industrial sensing platforms are not employed.
- Real-time data transmission primarily uses IoT communication protocols (e.g., LoRa as the main channel), enabling low-latency delivery of fuel and GPS data to the cloud backend.
- A GSM module functions as a fallback communication pathway, activated only during IoT connectivity loss, and is limited to transmitting essential alerts and critical status messages due to bandwidth and cost constraints.
- Other wireless communication technologies beyond the selected IoT and GSM modules are not implemented within the system.

c) **Limited Fleet Size and Geography**

- The system testing will be limited to the number of vehicles provided for use by the trucking company within a specific geographic area. Large scale testing and deployment across a wider range of fleet types and terrains are not covered.

d) **User Features**

- The cloud based dashboard is designed for the fleet managers use only and does not support interaction from the drivers.
- The cloud based dashboard is limited to essential features such as real time fuel level monitoring, GPS tracking, driver rest alerts, and maintenance reminders based on distance and fuel data. Advanced analytics features are not part of the current dashboard plans.

1.8 Description and Methodology

This study employs a quantitative research paradigm combined with a data-centric, system development approach to implement an AI-driven fleet fuel monitoring and anomaly detection system. The methodology integrates theoretical foundations from Chapter 3 on cyber-physical systems and IoT architectures, system design principles from Chapter 4, and practical implementation strategies detailed in Chapter 5. The system leverages real-time sensor data, including digitally calibrated float sensors interfaced via the vehicle's CAN bus or OBD-II port, GPS modules for location tracking, and embedded alert subsystems for driver safety. These components feed into a microcontroller that preprocesses signals, synchronizes time-series data, and transmits information to a cloud platform primarily via LoRa, with GSM as a backup channel.

Offline, historical and hybrid datasets, combining fleet telemetry and publicly available vehicle logs, are used to train non-parametric unsupervised machine learning models such as Matrix Profile and Isolation Forest. These models identify anomalous fuel consumption patterns and provide context-aware alerts. In the cloud, a Python-based engine continuously evaluates incoming data, logs anomaly events, and stores synchronized measurements in a centralized database. Users access the information through a web-based dashboard that displays real-time truck status, historical fuel usage, route playback, and operational alerts.

The methodology includes signal preprocessing (smoothing via moving averages), time synchronization using linear interpolation, hybrid dataset construction, and rigorous evaluation metrics (precision, false positive rate, accuracy, detection latency) to validate system objectives, as elaborated in Chapter 5. The approach follows iterative, agile development cycles, allowing continuous refinement of sensor integration, communication protocols, and anomaly detection algorithms, thereby ensuring reliability, scalability, and actionable fleet monitoring insights.

1.9 Estimated Work Schedule and Budget

1.9.1 Gantt Chart

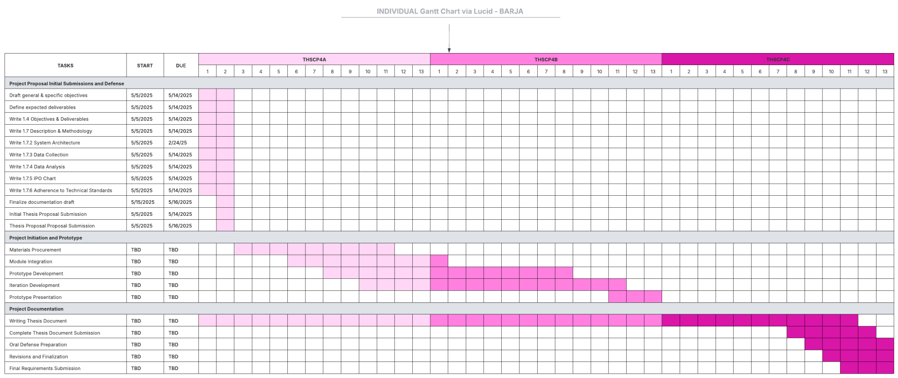

Fig. 1.1 Individul Gantt Chart for Barja

De La Salle University

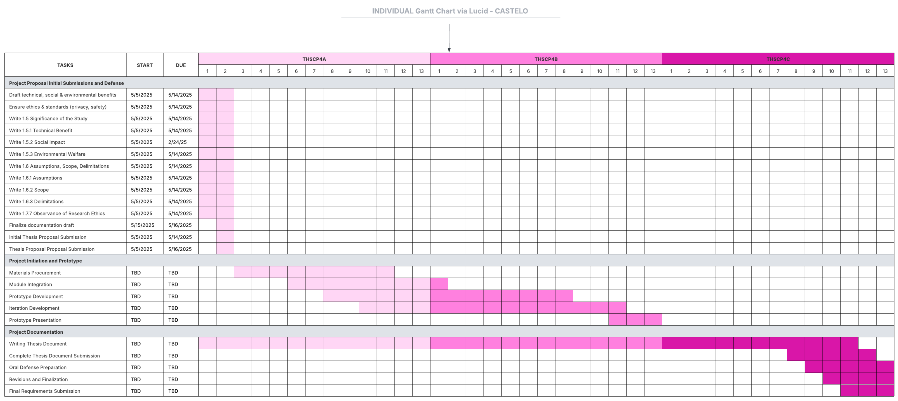

Fig. 1.2 Individul Gantt Chart for Castelo

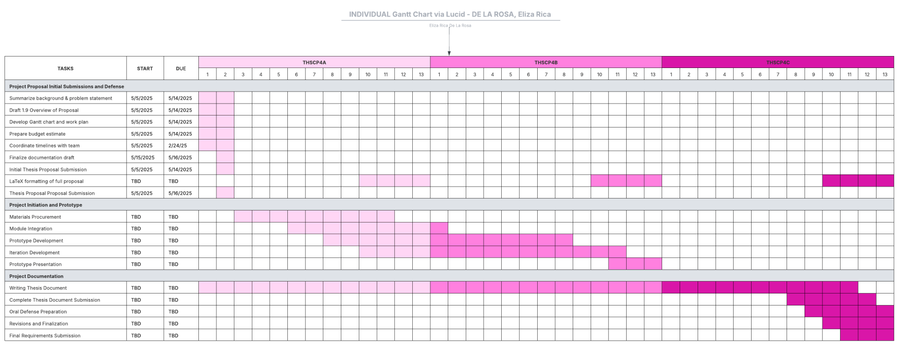

Fig. 1.3 Individul Gantt Chart for De La Rosa

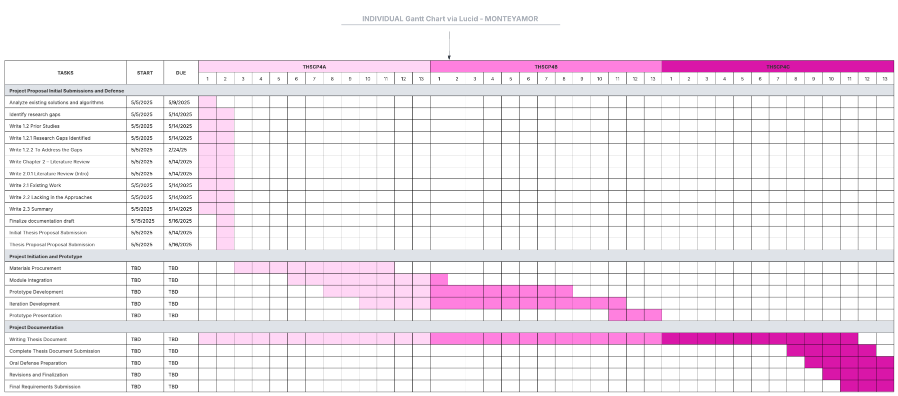

Fig. 1.4 Individul Gantt Chart for Montemayor

1.9.2 Estimated Budget

TABLE 1.3 DETAILED BUDGET BREAKDOWN FOR THE PROJECT

Item / Activity	**Unit**	**Qty**	**Unit Cost (PHP)**	**Total Cost (PHP)**	**Remarks**
A. Covered by Hino Motors					
Use of truck for testing	Truck	2	Sponsored	-	Provided by Hino
Fuel for testing runs	Tank fill	6	Sponsored	-	Provided by Hino
On-site sensor installation support	Lump sum	1	Sponsored	-	Technician/vehicle access
Access to maintenance data logs	Lump sum	1	Sponsored	-	Data-sharing support
B. Covered by Research Team					
MCU Development Board	-	1	1,800	1,800	Central controller for sensor interfacing
CAN Bus / OBD-II Interface Module	-	1	1,200	1,200	Reads fuel levels and vehicle parameters from ECU
GPS Module (5s logging)	-	1	650	650	Route tracking and speed monitoring
LoRa Transceiver Module	-	1	850	850	Primary IoT communication channel
GSM Module (SIM800L or similar)	-	1	900	900	Backup communication for critical alerts
Data SIM for GSM module	Monthly	3	200	600	For real-time fallback transmissions
Rest Alert Module (Buzzer + LED)	-	1	400	400	Driver fatigue / rest notification system
Breadboard, jumper wires, resistors, soldering tools	Lump sum	1	700	700	Hardware prototyping components
Transportation to Hino facility (testing/deployment)	Trip	6	250	1,500	Team visits for deployment and testing
Printing of documents and final reports	Pages	200	5	1,000	For panel copies and submission

Item / Activity	Unit	Qty	Unit Cost (PHP)	Total Cost (PHP)	Remarks
Binding of final manuscript	Copies	4	200	800	Panel and archive copies
Contingency Fund (10% of total)	Lump sum	1	-	970	For unforeseen expenses
Total Estimated Budget				**10,320**	

1.10 Overview of the Thesis Proposal

This chapter provides an overview of the study, which addresses inefficiencies in logistics and trucking operations while tackling safety concerns related to fuel monitoring, vehicle tracking, and anomaly detection. Modern trucking operations involve complex coordination, and current approaches—manual fuel measurements, fragmented GPS tracking, and delayed anomaly detection—are insufficient for ensuring operational efficiency and driver safety. This work proposes an integrated, IoT-based intelligent monitoring platform that leverages CAN bus fuel data, GPS, and cloud-deployed machine learning models to provide real-time insights and actionable alerts.

The problem statement highlights a fragmented monitoring environment with limited automation, reactive maintenance scheduling, and the inability to detect unusual fuel consumption proactively. The objectives of the research focus on developing a real-time system that reads fuel levels from the truck's ECU via CAN bus, tracks location and speed using GPS, and identifies anomalies in consumption patterns using unsupervised machine learning models (Matrix Profile and Isolation Forest). The system is supported by a cloud-based dashboard, which provides fleet managers with fuel usage insights, rest alerts, route tracking, and preventive maintenance notifications. This approach promotes operational efficiency, enhances driver safety, and reduces fuel wastage, making it a technically and environmentally relevant solution.

Building on this foundation, Chapter 2: Literature Review examines previous research on IoT-based fleet monitoring systems, GPS-enabled tracking technologies, and machine learning methods for anomaly detection. The review compares existing methods, identifies gaps in real-time deployment, system integration, and context-aware alerting, and evaluates technology innovations and algorithmic limitations. This analysis establishes the academic basis for the study and demonstrates how the proposed system advances the state of the art in smart fleet management by providing a fully integrated, real-time, and context-aware monitoring solution.

Chapter 2

LITERATURE REVIEW

2.0.1 Literature Review

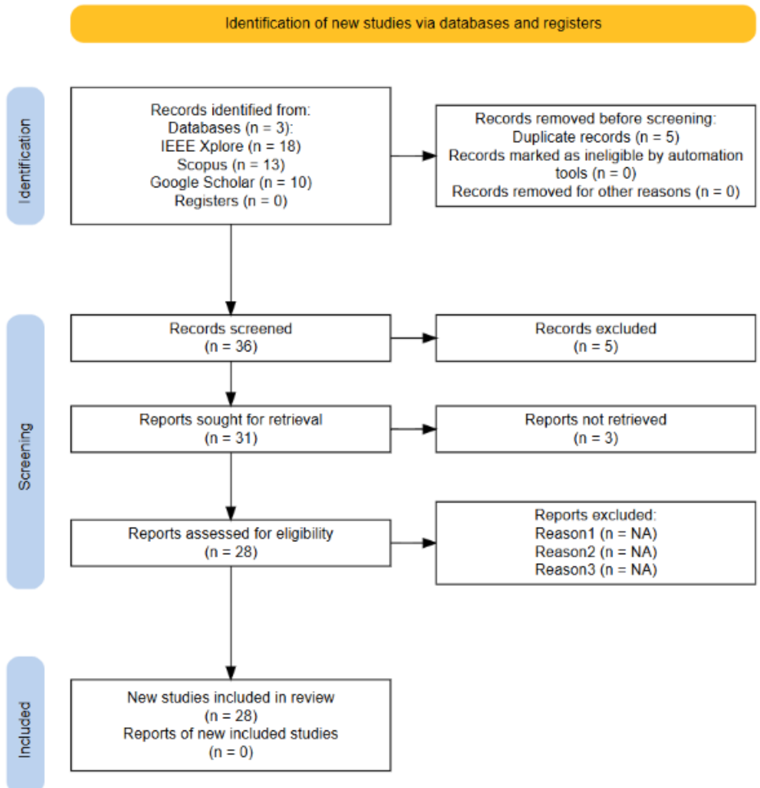

Fig. 2.1 PRISMA Diagram

The PRISMA diagram illustrates the systematic process used for selecting studies for review through screening and identification. The search identified a total of 41 records from three databases: Scopus, IEEE Xplore, and Google Scholar. Five (5) duplicate records were excluded from the 41 records, resulting in 36 records screened. The evaluation of 36 records resulted in five exclusions due to records being outdated/irrelevant, which reduced the number of reports needed for retrieval to 31. Three (3) reports were not retrieved upon retrieval, resulting in 28 reports assessed for eligibility. The final review included 28 studies only, while no new studies were reported.

2.1 Existing Work

2.1.1 Vehicle Tracking and Fuel Sensing Technologies

Fuel Monitoring Systems

Various fuel monitoring systems have been proposed, discussed, and developed in existing studies, emphasizing real-time fuel level, theft indication, and enhanced operational efficiency. In the work of Joshi et al. [Joshi, 2024], a system combines ESP32 and Arduino UNO microcontrollers with the ThingSpeak cloud-based platform to measure fuel levels, control transactions, and enable predictive maintenance at fuel stations. Similarly, Ahmed et al. [Ahmed et al., 2017] developed a fuel management system using Raspberry Pi and both ultrasonic and chemical Etape sensors to acquire real-time fuel level data, displayed through a web application.

In the automotive domain, several studies aim to enhance the accuracy of fuel management through sensor integration. The digital system by Sakthimohan et al. [M, 2021] employs ultrasonic and flow sensors with an Arduino UNO to display real-time measurements on an LCD, providing a replacement for imprecise analog gauges. Meanwhile, Farzana et al. [Farzana, 2020] proposed a motion-based fuel monitoring technique that integrates vehicle motion parameters with fuel level sensor readings, achieving 80% success in accurately determining fuel levels and 100% detection success in identifying fuel theft and refills via a smartphone application.

Studies focused on fuel theft detection, such as those by Tatababu et al. [Tatababu et al., 2024] and Mathi et al. [Mathi, 2024], utilize capacitive or ultrasonic sensors, GPS tracking, and GSM-based alerts to detect tampering or fuel loss and notify operators immediately. In the Intelligent Fuel Monitoring and Theft Detection System using GPS GSM, Mathi et al. [Mathi, 2024] also monitored battery charge (SoC), with applicability for both hybrid and conventional vehicles, making it suitable for logistics and fleet management.

Ali et al. [Komal Shoukat Ali, 2018] presented an automatic fuel tank monitoring, tracking, and theft detection system integrating Arduino-based fuel circuits with GSM/GPS modules and LabVIEW for database management. This system replaces manual fuel tracking with automated alerts and volume monitoring, offering a cost-effective solution for transportation industries. Overall, these systems indicate a trend toward inexpensive, sensor-based, remotely accessible fuel monitoring technologies with high priority on accuracy, security, and real-time data analysis.

GPS and Vehicle Telematics

Global Positioning System (GPS) and vehicle telematics technologies have become essential components in modern vehicle monitoring and fleet management systems. These technologies allow real-time tracking, remote diagnostics, and performance analysis by collecting and reporting data such as location, speed, fuel consumption, and engine status. A common trend in these studies is the combination of GPS and GSM modules to facilitate real-time vehicle tracking, usually complemented by cloud-based or mobile-supported platforms. Joshi et al. [Joshi, 2024] proposed a system enabling real-time monitoring for logistic management using sensors, GPS tracking, and cloud-based data analytics. Similarly, Alquhali et al. [Alquhali et al., 2019] and Nada [Nada, 2017] employed GPS and GSM modules to send live location data, show vehicle tracks on electronic maps, and record location history for future reference and analysis. Vigneshwaran et al. [Vigneshwaran et al., 2020] presented a GPS-GSM-based bus tracking system supporting student notifications for bus arrival, emergency notifications, and route alerts, enhancing safety and saving waiting time. The system further offered a framework for real-time fuel monitoring with implications for broader fleet management applications.

2.1.2 Anomaly Detection

Several studies on anomaly detection in vehicular systems aim to improve operational safety, combat theft and fraud, and optimize fuel efficiency. Various technological means, such as GPS tracking, machine learning, and time series analysis, have been used to identify abnormal vehicle behaviors. Crisgar et al. [Crisgar, 2021] developed a vehicle tracking system using Google Cloud IoT Core and Firebase to monitor vehicle movements and engine activity for theft detection. Unauthorized ignition or unexpected movement beyond a 50-meter radius is flagged as potential theft, with real-time notifications pushed to a Progressive Web App, enabling users to respond quickly.

Studies focusing on fuel efficiency in heavy-duty vehicles (HDVs) through anomaly detection models include Turan et al. [Turan et al., 2024], who proposed an ensemble of unsupervised machine-learning models (Isolation Forest, Autoencoder, k-NN Regression) combined via weighted majority voting to detect abnormal fuel consumption patterns based on road slope and speed variations. Mumcuoglu et al.

[Mumcuoglu, 2023] used decision tree ensembles to classify fuel usage as normal or anomalous, achieving 92.2% accuracy and an F1 score of 0.78. Both studies indicate that repeated detection of outliers can highlight anomalies such as mechanical issues or inefficient driving.

Aquize et al. [Aquize et al., 2017] applied Self-Organizing Maps (SOM) to detect abnormal fuel consumption in RFID-registered vehicles in Bolivia. By learning conventional consumption patterns, the SOM model achieved an 80% confidence rate in flagging suspicious activity indicative of fraud. Semenov et al. [Semenov et al., 2024] employed DBSCAN clustering and Local Outlier Factor (LOF) in a three-level telematics platform to forecast and detect truck breakdowns in real-time. This system tracks multiple vehicle parameters, labeling them as normal or outliers based on deviation from learned patterns, minimizing surprise breakdowns and providing early driver alerts.

Garcia Vega [Garcia Vega, 2022] emphasized time series anomaly detection for vehicle monitoring, developing an iterative approach to detect unusual driving patterns through preprocessing, subsequence segmentation, scoring, clustering, and expert validation. Applied in an industrial context, it identified over 120 validated anomalies, helping diagnose mechanical failures or unsafe driving behavior. The study highlights the importance of visual interpretability tools and recommends future improvements via real-time data processing.

2.1.3 Machine Learning Applications

Recent advances in machine learning have been integrated into vehicle systems to improve fuel efficiency, detect anomalies, and enhance operational transparency. In fuel monitoring and anomaly detection, machine learning methodologies extract actionable insights from complex, high-dimensional vehicle data. Fortela et al. [Fortela, 2023] applied unsupervised machine learning algorithms (Principal Components Analysis, K-Means Clustering, Self-Organizing Maps) to detect anomalies in fuel economy and emission data from light-duty vehicles (2015–2023). These methods revealed clusters where CO emissions were positively linked to fuel economy, contrary to expected patterns, highlighting irregularities caused by specific fuel test procedures. Barbado and Corcho [Barbado and Corcho, 2022] combined unsupervised anomaly detection with domain knowledge and interpretable machine learning (XAI) to explain potential causes of fuel consumption anomalies. Their models demonstrated up to 88% potential fuel savings on larger datasets and across various user roles.

2.2 Lacking in the Approaches

While existing technologies in vehicle tracking, fuel sensing, and anomaly detection form the foundation upon which developments can be built, they suffer from critical limitations that prevent their use in real-time, fleet-wide operations. Research works such as Krishnasamy [Krishnasamy, 2019] have employed basic sensor technologies without any validation of fuel measurements in real-time driving environments or estimation of accuracy parameters relative to manual readings. Besides, current

TABLE 2.1 SUMMARY OF EXISTING WORKS

Study	Technology Used	Key Features	Strengths	Limitations
Tatababu et al. (2024) [Tatababu et al., 2024]	Ultrasonic/Capacitive sensors, GSM, GPS, Smartphone/Web interface	Theft detection, fuel level alert, real-time remote access	Accurate for remote generator monitoring, GPS-based tracking	Findings specific to the tested generator systems and environments
Joshi et al. (2024) [Joshi, 2024]	IoT, GPS, RFID, Cloud Computing	Real-time vehicle/asset tracking, data analytics, cloud-based monitoring	High tracking accuracy, improved logistics efficiency, responsive alert system	Limited by hardware and logistics test case set
Vigneshwaran et al. (2020) [Vigneshwaran et al., 2020]	GPS, GSM	Real-time bus tracking, emergency alert system, communication to students, route updates	Improves safety, scheduling, and passenger information	Not highly scalable; lacks detailed integration with modern telematics platforms
Patil & Patil (2024) [Patil and Patil, 2024]	IoT, GPS, GSM, Accelerometer, IR Sensors	Tracking system with SMS-based communication; remote ignition cut-off mechanism and vehicle password protection	Comprehensive vehicle security, real-time alerts, detection of tampering/impacts	Low adaptability to harsh conditions like GPS/GSM signal loss in remote areas
Fortela et al. (2023) [Fortela, 2023]	Unsupervised Machine Learning: PCA, K-Means, SOM	Detects hidden trends and impending anomalies; analyzes correlation between CO emissions and fuel economy	Effectively identifies non-obvious patterns within clusters; highlights flaws in standardized testing procedures	Lacks integration with real-time or operational vehicle data; conclusions depend on historical datasets
Barbado & Corcho (2022) [Barbado and Corcho, 2022]	Unsupervised Anomaly Detection; Explainable Boosting Machine (EBM)	Provides fuel anomaly detection and cause explanations, tailored recommendations for fleet operators	High interpretability via XAI metrics; role-based recommendations; potential fuel savings up to 88%	Performance depends heavily on quality of telematics data or feature relevance
Aquize et al. (2017) [Aquize et al., 2017]	Self-Organizing Maps (SOM)	Detects fraudulent fuel usage based on RFID data patterns	80% accuracy applied to real socio-economic problem	Used only simple SOM model; accuracy can improve with more context variables

systems are limited to specific uses and are not tested for scalability or integration with fleet vehicles like trucks such as institutional/organizational level of motorcycles in the study of RS et al. [RS et al., 2024]. Some works do consider machine learning techniques for anomaly detection, but such techniques are mostly based on static and post-processed datasets, with no real-time on-vehicle analysis. Very few implementations include unsupervised machine learning and advanced context-aware modeling, and precision benchmarks are usually left unreported, if at all addressed.

Ultimately, there is no prior system that provides a full solution that brings together a cloud-connected solution for real-time fuel monitoring with integration of GPS tracking and anomaly detection. Further, it completely lacks features such as automatic rest alerts and predictive maintenance reminders, which are necessary for safe, efficient, and compliant operations. Such shortcomings create an opportunity for a more integrated, intelligent, and real-time solution, and this is what the thesis proposes to address.

2.3 Summary

With this chapter, it was discovered that there have been various studies on vehicle tracking, fuel sensing, and anomaly detection. The PRISMA diagram shows the systematic approach that was used in the identification of the studies, out of which the 41 initial records were narrowed down to 28 after excluding the duplicates, outdated/irrelevant, and not retrieved records. Moreover, the existing studies presented that though a lot has been done to develop individual systems for fuel level monitoring, GPS tracking, and anomaly detection, most of these works are either narrowly

TABLE 2.2 IDENTIFIED GAPS, LIMITATIONS, AND PROPOSED SOLUTIONS

Gap Identified	**Limitations**	**Proposed Solution**
Area of precision	Existing models lack context-awareness and show lower precision. Models presented in the study of Barbado & Corcho [Barbado and Corcho, 2022] performed up to 80% of anomalous instances explained.	(SO4) To design and evaluate a context-aware fuel usage deviation modeling approach using non-parametric unsupervised machine learning models, like Matrix Profile and Isolation Forest, achieving ≥90% precision and ≤10% false positives in identifying anomalous fuel consumption patterns within controlled environments.
Absence of a comprehensive system integrating fuel anomaly detection with GPS-based tracking, and predictive maintenance and alerts	Previous works, including those by Joshi et al. [Joshi, 2024] and Nada [Nada, 2017], focus on real-time tracking using GPS/GSM, but lack integration with anomaly detection methods for analyzing fuel consumption patterns or predicting vehicle maintenance issues.	(SO3, SO4, SO5) To capture and log GPS location data via an embedded module, correlating route information with fuel and time data for behavior tracking; design and evaluate a context-aware fuel usage deviation modeling approach using non-parametric unsupervised machine learning models; develop and integrate rest alerts and maintenance reminders into a cloud-based dashboard.

focused or operate in isolation. Many rely on basic sensor integrations without advanced analytics, or implement machine learning models without connecting them to real-time, in-vehicle data streams.

Numerous studies have exhibited the application of different sensor-based systems for fuel monitoring in real-time and detecting theft, in many cases involving microcontrollers, GSM, and GPS modules. Likewise, research on vehicle telematics illustrates the application of GPS in route tracking and fleet management. In the field of anomaly detection, some methods based on unsupervised machine learning and clustering techniques have been utilized to detect abnormal fuel consumption patterns and flag fraudulent or inefficient activity. Yet, most of these researches are based on special scenarios, are not integrated with real-time operational data, or

De La Salle University

depend on historical data.

While advances have been made in each of these domains, a significant void exists in creating a holistic, real-time solution that integrates fuel level measurement, GPS tracking, and intelligent anomaly detection through machine learning. Current systems exist in silos, are vehicle type- or use case-specific, and do not commonly incorporate predictive maintenance or driver feedback mechanisms.

This research seeks to fill these gaps by designing and testing an integrated IoT-based system tailored for trucks. The system will be an integrated system that combines real-time fuel monitoring, GPS tracking, and advanced anomaly detection using machine learning which aims to improve operational efficiency, driver safety, and maintenance planning.

Chapter 3

THEORETICAL CONSIDERATIONS

This chapter establishes the theoretical foundations underpinning the proposed IoT-based fleet fuel monitoring and anomaly detection system. It presents the essential concepts, mathematical models, and scientific principles that guide the development of the system. These theories act as the main foundation of the study, supporting the methodological decisions described in Chapter 5. By understanding the underlying theories—ranging from cyber-physical systems to signal processing and machine learning—the relevance and rigor of the chosen approach become clear.

3.1 Overview

The methodologies used in the system are grounded in established theories in embedded systems, time-series processing, IoT architectures, unsupervised machine learning, and vehicular telematics. These theoretical concepts justify the design choices made in the development of real-time fuel monitoring components, GPS-based tracking, and anomaly detection models. This chapter explains the foundations behind the signal models, communication processes, and machine learning algorithms that make accurate and robust system performance possible.

3.2 Theoretical Foundations

3.2.1 Cyber-Physical Systems (CPS)

The proposed solution functions as a Cyber-Physical System, integrating embedded sensors, microcontroller-based processing, cloud computation, and real-time communication. CPS theory emphasizes:

- the tight coupling between physical signals and digital processing,
- deterministic timing and synchronized data flow,
- continuous monitoring and computational feedback loops.

These principles justify the system's architecture, where fuel-level signals, GPS data, and operational states are processed and transmitted in real time.

3.2.2 Sensor Theory and Signal Measurement

Fuel-level readings extracted through the CAN bus/OBD-II interface follow digital measurement theory. Sensor signals can exhibit noise, quantization error, and environmental fluctuation. To improve stability, the system uses a Simple Moving Average (SMA) filter, a standard smoothing method defined as:

$$x_t^{\text{smooth}} = \frac{1}{N} \sum_{i=0}^{N-1} x_{t-i} \tag{3.1}$$

This reduces high-frequency noise and creates a clearer trend for anomaly detection algorithms.

3.2.3 Time-Series Synchronization Theory

Sensor streams often arrive at irregular timestamps. Synchronizing them requires interpolation theory. For two samples at timestamps T_i and T_{i+1}, the interpolated point at t_j is computed as:

$$x_j = x_i + \left(\frac{x_{i+1} - x_i}{T_{i+1} - T_i} \right) (t_j - T_i) \tag{3.2}$$

This ensures aligned and consistent time-series data across fuel level, GPS speed, and distance.

3.2.4 IoT Communication Theory

The communication subsystem relies on Internet of Things (IoT) principles and long-range wireless communication theory. LoRa operates using Chirp Spread Spectrum (CSS) modulation, characterized by:

- long-range transmission,
- low power consumption,
- robustness to interference.

Theoretical latency between sender and cloud receiver is expressed as:

$$L = T_{\text{received}} - T_{\text{sent}} \tag{3.3}$$

Understanding latency is essential for analyzing responsiveness and real-time performance.

3.2.5 GPS Temporal and Spatial Theory

GPS logging follows Global Navigation Satellite System (GNSS) principles. To evaluate consistency of GPS sampling, the inter-arrival interval is computed as:

$$t_i = T_i - T_{i-1} \tag{3.4}$$

Maintaining a stable interval ensures accurate reconstruction of routes and distance traveled.

3.2.6 Unsupervised Anomaly Detection Theory

The anomaly detection component relies on non-parametric, unsupervised machine learning. These methods identify unusual behaviors in multivariate time-series data without labeled training samples.

3.2.6.1 Matrix Profile Theory

Matrix Profile computes similarity between subsequences within a time series. For a time series T of length n and window size m, the profile value is:

$$P[i] = \min_{j \neq i} \|T_{i:i+m-1} - T_{j:j+m-1}\| \tag{3.5}$$

The subsequences with the highest $P[i]$ values, known as discords, are likely anomalies. This method is ideal for detecting sudden fuel drops or siphoning patterns.

3.2.6.2 Isolation Forest Theory

Isolation Forest isolates anomalies through random partitioning. An anomaly score is expressed as:

$$s(x) = 2^{-\frac{E(h(x))}{c(n)}} \tag{3.6}$$

Where:

- $E(h(x))$ is the expected path length for observation x,
- $c(n)$ normalizes the tree depth by dataset size.

Higher scores indicate higher likelihood of anomalous fuel behavior.

3.3 Summary

This chapter presented the theoretical foundation for the system's methodology. It discussed the key concepts supporting real-time sensing, signal preprocessing, GPS logging, IoT communication, and unsupervised anomaly detection. By establishing these theoretical pillars, this chapter clarifies the scientific basis that makes the system's design viable, accurate, and aligned with established models in engineering and data science.

Chapter 4

DESIGN CONSIDERATIONS

This chapter presents the design considerations that guided the development of the proposed IoT- and AI-driven fuel monitoring and anomaly detection system. Whereas Chapter 3 established the theoretical foundations of IoT architectures, cyber-physical systems, signal processing, and unsupervised anomaly detection, the present chapter translates those theories into practical engineering decisions that were directly implemented in Chapter 5.

These design considerations serve as the structural “columns” of the system: grounded in accepted standards, ethical guidelines, and engineering best practices, but tailored to support the procedural steps carried out in Chapter 5. The discussion begins with a comprehensive account of the technical standards that ensure reliability, safety, interoperability, and data security, followed by software and hardware design decisions. Ethical considerations—including privacy, responsible AI, and driver safety—are also presented.

4.1 Standards

4.1.1 Adherence to Technical Standards

To ensure reliability, accuracy, and safety, the proposed fuel anomaly detection system is developed in accordance with established engineering and IT standards. These standards ensure interoperability, scalability, and long-term viability. They directly influence the system architecture, data handling, and model implementation workflows outlined in Chapter 5.

a) **Standards for Sensors and Hardware**

The system integrates sensors and modules that employ standard communication protocols—**CAN bus/OBD-II, UART, I2C, and GPIO**—consistent with IEEE 1451 smart transducer guidelines [IEEE Standards Association,]. The CAN bus/OBD-II fuel data acquisition aligns with industry telematics conventions [iso, 2020], while the GPS module adheres to global navigation and timing standards [NASA EarthData, 2025].

Safety-related components (buzzer, indicators, MCU GPIO outputs) comply with IEC 61131-2 and ISO 7637 automotive power and electromagnetic compatibility guidelines [iec, , iso, d]. All embedded electronics meet CE and FCC certifications [cec, 2022, fcc,] to ensure environmental and electrical safety.

b) **Data Acquisition and Signal Processing**

Signal conditioning and acquisition follow recommended standards for reliable sensor data handling [iso, e]. A Simple Moving Average (SMA) filter is applied to the fuel-level signal to reduce noise while preserving sharp variations indicative of anomalies.

Time alignment of fuel, GPS, speed, and engine readings follows ISO/IEC 19510 principles for structured time-series data [iso, e]. These standard-aligned preprocessing steps feed into the anomaly detection models described in Chapter 5.

c) **Communication and Data Transmission**

Communication follows TCP/IP standards under IETF and IEEE 802.11 [rfc, , Samainstage, 2025], while MQTT serves as the primary IoT protocol following OASIS specifications. End-to-end encryption via TLS/SSL aligns with ISO/IEC 27001 cybersecurity requirements [iso, 2022a].

These communication strategies form the basis for the system's message-handling and cloud-upload mechanisms in Chapter 5.

d) **Machine Learning and Model Validation**

The Matrix Profile and Isolation Forest models are implemented using open-source libraries, and their evaluation follows ISO/IEC 29119 software quality standards [iso, 2022b]. Validation metrics—confusion matrices, ROC curves, precision–recall graphs—provide statistical grounding for model evaluation, as presented in Chapter 5.

e) **Software Architecture and Data Management**

The modular software design complies with ISO/IEC/IEEE 42010 architectural principles [iso, g]. ACID-compliant storage and GDPR-/ISO 27018-aligned data protection protocols [iso, f] ensure data integrity and privacy. All processed datasets follow FAIR principles.

4.1.2 Observance of Research Ethics

Ethical considerations guide the responsible deployment of the system, especially given the safety-critical and privacy-sensitive context of transportation fleets. Three major ethical domains were identified: data privacy, system safety, and responsible AI.

Data Privacy and Security

The system adopts:

- informed consent prior to any data collection,

- strict access control and secure data storage,
- anonymization and aggregation when feasible,
- transparency in data usage and access rights,
- compliance with national and international privacy regulations.

System Safety and Occupational Risk Management

Given the operational risks of fleet environments, the design aligns with ISO 45001, ISO 26262, and ISO 39001 [iso, c, iso, a, iso, b]. Safety-oriented measures include:

- minimizing driver distraction,
- reducing false alarms through validation,
- providing adequate training for users,
- implementing clear protocols for anomaly response.

Responsible AI and Algorithmic Accountability

Consistent with IEEE P7003 [iee,], the system incorporates:

- explainability mechanisms for anomaly outputs,
- documentation of roles and responsibilities in alert handling,
- defined liability procedures for system errors,
- safeguards against algorithmic bias.

4.2 Technical Design Considerations

This section elaborates on the engineering decisions and standards that guided the development of the proposed IoT- and AI-driven fuel monitoring and anomaly detection system. Each component, software layer, algorithm, and communication protocol is grounded in established standards to ensure reliability, safety, interoperability, and security. The references indicate where each design decision is applied in the implementation procedures described in Chapter 5.

4.2.1 Hardware Selection and Integration

The hardware configuration is designed to support the end-to-end workflow of the system (see Chapter 5.2). The following standards and implementation points apply:

- **MCU:** ESP32/NodeMCU serves as the edge processing device for real-time acquisition and preprocessing of fuel, GPS, and vehicle telemetry data. It complies with IEC 61131-2 for embedded controllers [iec,] and ISO 7637 for automotive electrical transients [iso, d]. In Chapter 5.2, the MCU handles sensor filtering, timestamping, and local anomaly scoring.
- **Fuel Sensor:** A resistive fuel-level sensor provides analog voltage proportional to the fuel level. The sensor integration aligns with IEEE 1451 smart transducer communication standards [IEEE Standards Association,]. Chapter 5.2 describes the calibration process, analog-to-digital conversion, and SMA filtering of sensor data.
- **GPS Module:** The UART-connected GPS module provides geolocation and speed data, adhering to global navigation standards [NASA EarthData, 2025].

Its signals are synchronized with fuel readings during preprocessing in Chapter 5.2.

- **CAN/OBD-II Interface:** Vehicle telemetry (engine RPM, ignition status, and ECU-reported fuel consumption) is captured using the CAN/OBD-II interface, following ISO 15765 and ISO 15031 telematics standards [iso, 2020]. Chapter 5.2 details how this data is collected and correlated with sensor readings.
- **Safety Indicators:** Buzzer and LED outputs comply with IEC 61131-2 and ISO 26262 functional safety requirements [iso, a]. These indicators provide immediate alerts to drivers (Chapter 5.2, Subsection on Driver Alerts).

4.2.2 Software Architecture

The modular software layers are designed according to ISO/IEC/IEEE 42010 software architecture standards [iso, g], supporting scalability, maintainability, and interoperability. Each layer's function is mapped to its implementation in Chapter 5.2:

- **Data Acquisition Layer:** Captures sensor and telemetry data, applies SMA and median filters, and performs anomaly threshold checks. Implementation details are in Chapter 5.2.
- **Processing Layer:** Performs edge anomaly detection, timestamping, and data aggregation. Standards for real-time processing and task scheduling ensure predictable latency. Implementation is in Chapter 5.2.
- **Cloud Integration Layer:** MQTT-based JSON payloads are used for cloud ingestion, following OASIS MQTT v5 and ISO/IEC 27001 security standards [iso, 2022a]. TLS/SSL ensures encrypted transmission (Chapter 5.2.

De La Salle University

- **User Interface Layer:** Dashboards display real-time fuel levels, GPS locations, anomaly alerts, and historical trends. Web standards (HTML5, CSS3, JavaScript) and secure REST APIs are employed, as described in Chapter 5.2.

4.2.3 Machine Learning and Algorithms

Machine learning components follow ISO/IEC 29119 software quality standards [iso, 2022b] and IEEE P7003 AI accountability guidelines [iee,]:

- **Models:** Isolation Forest and Matrix Profile detect anomalies both at edge and cloud levels. Their implementation and evaluation are detailed in Chapter 5.2.
- **Evaluation Metrics:** Confusion matrices, ROC curves, and precision–recall graphs are used for model validation, ensuring alignment with ISO/IEC 25010 software quality characteristics. Chapter 5.2, Chapter 5.3 provides practical application in performance evaluation.
- **Alternative Models:** Autoencoders and LSTM networks were considered but excluded due to edge processing constraints, as justified in Chapter 4.3.

4.2.4 Communication and Security

Communication protocols and security measures are aligned with ISO/IEC 27018, ISO/IEC 27001, and GDPR standards to ensure data privacy, encryption, and secure access control:

- **MQTT Protocol:** Supports lightweight, reliable IoT messaging, implemented with QoS 1 to guarantee delivery. Applied in Chapter 5.2.

- **TLS/SSL Encryption:** Provides end-to-end encryption for transmitted data. Compliance with ISO/IEC 27001 ensures data integrity and confidentiality. Applied in Chapter 5.2.
- **Role-Based Access and Anonymization:** Enforces secure data access policies and anonymizes sensitive driver information, complying with ISO/IEC 27018 and GDPR principles. Applied in Chapter 5.2.
- **JSON Serialization:** Standardized data format for cloud compatibility, ensuring interoperability with cloud services and dashboards. Applied in Chapter 5.2.

4.3 Alternative Software Designs and Cost Considerations

During the planning and prototype development phase, several candidate software designs were evaluated for their suitability to the system's requirements. Table 4.1 summarizes the strengths, potential risks, and fit relative to specifications. The hybrid Matrix Profile + Isolation Forest (MP + IF) configuration was selected for implementation and used during prototype testing due to its balance of accuracy, computational efficiency, and anomaly coverage.

A cost study was conducted after prototype testing to confirm that the system configuration was cost-effective. This study reviewed hardware, software, and development resources, comparing the chosen open-source and local execution setup against alternative configurations such as commercial analytics tools, cloud-based computing, or hardware-integrated telemetry systems.

TABLE 4.1 ALTERNATIVE SOFTWARE DESIGNS EVALUATED FOR THE PROTOTYPE

Candidate	Strengths	Risks / Trade-offs	Fit vs. Spec
Isolation Forest (IF)	Fast, unsupervised, handles multivariate data; simple to deploy using scikit-learn	Limited temporal context; sensitivity to contamination parameter may affect false alarms	High
Matrix Profile (MP)	Shape-based time-series discord detection; ideal for sudden fuel drops or irregular usage patterns; efficient in local runtime	Requires window-size tuning; primarily one-dimensional unless extended for multivariate data	Very High
Clustering (KMeans / DBSCAN)	Captures collective anomalies (trip-level patterns) and provides intuitive grouping for fleet behavior	Requires feature engineering; parameter sensitivity can reduce consistency	Medium
Hybrid (MP + IF / IF + LOF)	Combines time-series and statistical strengths; reduces false positives and broadens anomaly coverage	Higher computational cost and added complexity in fusion or ensemble scoring	Very High
Deep Learning (LSTM / TCN)	Learns complex multivariate temporal relationships and adapts well with large real-world datasets	Data-hungry, less interpretable, requires more computing resources for training and deployment	Medium

The total estimated budget for developing and implementing the prototype system is **PHP 10,320**, confirming that the selected configuration remains the optimal and sustainable choice for achieving the project's objectives.

4.4 Summary

This chapter presented the key technical and ethical design considerations that shape the system. By grounding the project in ISO, IEEE, OASIS, GDPR, and FAIR standards, the system ensures reliable data acquisition, secure communication, validated machine learning, and responsible AI use.

These design elements establish the foundation for the procedures described in Chapter 5, bridging theoretical concepts from Chapter 3 with the practical implementation steps of the system.

Chapter 5

METHODOLOGY

This chapter presents the research methodology employed for the development of an AI-driven fuel monitoring and anomaly detection system for fleet vehicles. The methodology integrates a multi-layered approach combining sensor-level data acquisition, edge computing, IoT communication, cloud-based machine learning, and client-side visualization. The approach emphasizes real-time anomaly detection, accurate fuel monitoring, operational transparency, and actionable driver and fleet management insights.

The research methodology is structured to address both general and specific objectives, ensuring that the system is technically feasible, scalable, and reliable. Figure 5.2 and Table 5.1 summarize the methods and approaches for each objective.

TABLE 5.1 SUMMARY OF METHODS FOR REACHING THE OBJECTIVES

Objectives	**Methods / Approaches**	**Locations**

GO: Develop and evaluate an IoT-based system for trucks in collaboration with Hino Parañaque Subic GS Auto Inc. (SGSA) that monitors real-time fuel levels, tracks routes using GPS, and detects unusual fuel consumption using machine learning, improving driver safety, supporting rest scheduling, and facilitating proactive vehicle maintenance through automated alerts.	• Embedded MCU preprocessing and aggregation • LoRa/GSM communication for low-latency and fail-safe data transfer • Cloud-based machine learning for anomaly detection • Web dashboard for visualization • Driver alert subsystem for operational safety	Sec. 5.2
SO1: Measure fuel levels using digital float sensors integrated through the truck's ECU/diagnostic connector, achieving ±5% accuracy compared to manual readings in controlled conditions.	• Digital fuel sensors interfaced with ECU/OBD-II • Signal filtering and smoothing • Calibration against manual measurements • Continuous validation for measurement reliability	Sec. 5.2

SO2: Transmit real-time fuel and GPS data through IoT-based communication modules to a cloud server and central web dashboard with latency not exceeding 3 seconds.	• Real-time IoT transmission via LoRa (primary) and GSM (backup) • Cloud ingestion APIs • Latency measurement and benchmarking to ensure timely reporting	Sec. 5.2
SO3: Capture and log GPS location data every 5 seconds via an embedded module, correlating route information with fuel and time data for behavior tracking.	• GPS-based location logging at 5-second intervals • Synchronization with fuel data • Structured cloud database for multi-source data storage and route–fuel correlation	Sec. 5.2
SO4: Design and evaluate a context-aware fuel usage deviation modeling approach using non-parametric unsupervised ML models (Matrix Profile or Isolation Forest), achieving 90% precision and 10% false positives.	• Offline training of anomaly detection models using Isolation Forest and Matrix Profile • Evaluation using precision, FPR, and accuracy • Context-aware thresholding to reduce false positives	Sec. 5.2

SO5: Develop and integrate hardware-based rest alerts and maintenance reminders for long-haul driver safety linked to a cloud-based dashboard hosted at the partner trucking company's base, using cumulative fuel and distance data to guide break schedules and service intervals (300 km or 6-hour rest recommendations, 5,000 km oil check reminders).	• Driver rest and maintenance alert system with cumulative distance/time tracking • Cloud-based monitoring and logging • Alert notification via dashboard and in-vehicle signals	Sec. 5.2

5.1 System Architecture and Conceptual Design

The system follows a layered architecture with a focus on modularity, scalability, and reliability. The conceptual framework (Figure 5.1) integrates the following:

- **Vehicle Sensing Subsystem:** Digitally calibrated fuel-level sensors, GPS modules, and driver rest-time monitors. Sensor signals are collected via ECU or OBD-II ports.
- **Edge Processing:** The MCU performs signal preprocessing, filtering, smoothing, synchronization, and timestamping before transmission.
- **IoT Communication:** LoRa transceivers enable low-power, long-range data transmission; GSM acts as a backup channel for essential alerts, balancing cost

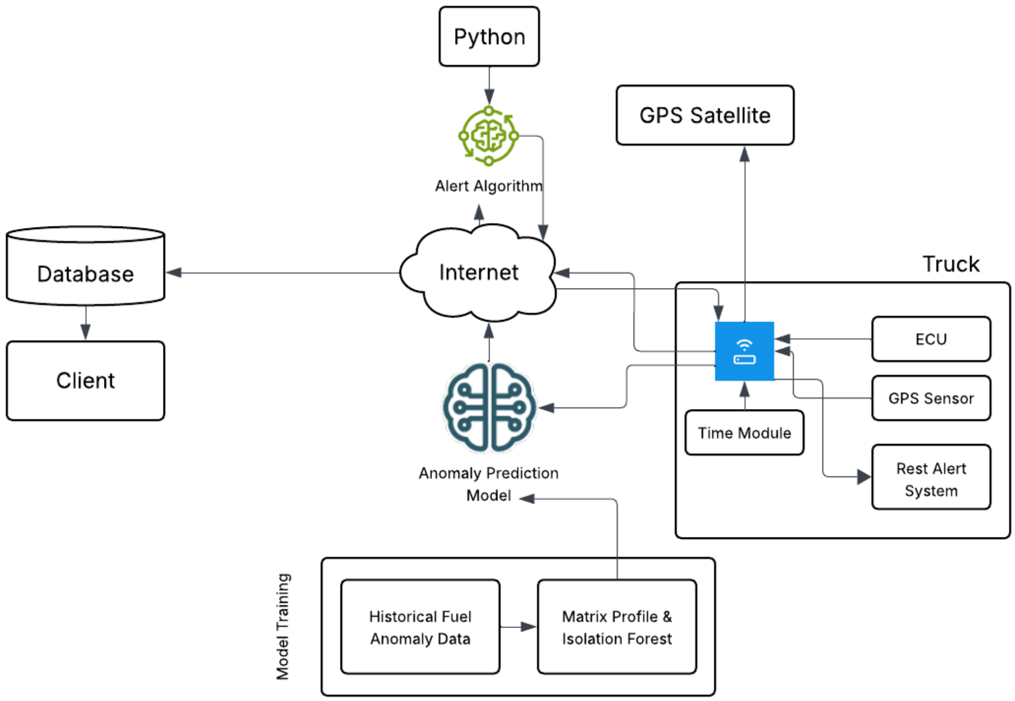

Fig. 5.1 Proposed Conceptual Diagram

and bandwidth considerations.

- **Cloud Intelligence:** Historical data is used to train unsupervised ML models (Isolation Forest, Matrix Profile) to detect anomalies. The cloud evaluates incoming real-time data and triggers alerts.
- **Client Visualization:** A web-based dashboard provides real-time visualization of fuel levels, routes, alerts, and historical playback for decision-making.

The system architecture (Figure 5.2) follows a layered, end-to-end IoT design composed of three major layers: the data acquisition layer, the communication and cloud processing layer, and the data visualization layer. This structure supports real-time

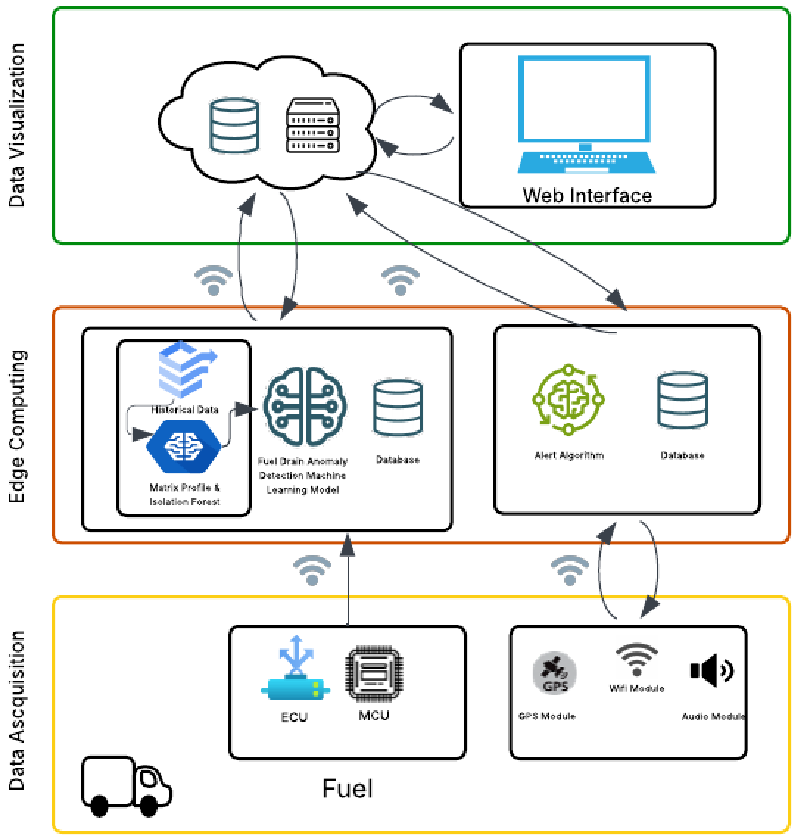

Fig. 5.2 Proposed System Architecture Diagram

fuel monitoring, GPS-based route tracking, driver safety alerts, and cloud-based anomaly detection.

At the data acquisition layer, the vehicle is outfitted with digitally calibrated float sensors integrated through the truck's ECU/diagnostic connector to obtain accurate fuel-level measurements. The microcontroller (MCU) serves as the primary edge device, interfacing with the float sensor, GPS module (logging coordinates at 5-second intervals), rest alert subsystem, and time-tracking module. A LoRa transceiver functions as the main channel for transmitting fuel, location, and operating-hour data to the cloud, while a GSM module is used strictly as a fallback communication interface, activated only when LoRa connectivity is unavailable and limited to essential alerts due to bandwidth and cost constraints.

The communication and cloud processing layer handles real-time data ingestion and anomaly detection. Incoming packets from the vehicle are received through cloud-based API endpoints and routed to processing services where the trained machine learning model—developed offline using Matrix Profile or Isolation Forest—is deployed for fuel anomaly evaluation. A Python-based alert engine evaluates fuel–location–time patterns and generates automated rest reminders, maintenance notifications, and anomaly flags. All validated data and event logs are stored in a centralized cloud database for long-term analysis.

The data visualization layer consists of a web-based monitoring dashboard that retrieves information from the cloud database. Through this interface, clients can view real-time truck locations, fuel levels, distance traveled, driver operating hours, and alert notifications. The dashboard also supports historical playback of route and fuel usage trends, enabling improved operational awareness and proactive decision-making.

Overall, the architecture highlights the complete data flow—from ECU-integrated sensing and MCU preprocessing, to LoRa/GSM communication, to cloud-based machine learning and dashboard visualization—demonstrating a scalable and modular IoT system for fleet monitoring and anomaly detection.

5.2 Implementation

5.2.1 Data Acquisition and Preprocessing

Data acquisition and preprocessing are crucial for reliable anomaly detection. The methodology includes:

a) **Sensor Calibration:** Fuel-level sensors are calibrated using known volumes, verified with manual measurements, and cross-checked with historical data trends.

b) **Signal Filtering:** Raw sensor signals are filtered using moving-average and median filters to reduce noise caused by vehicle motion or fuel sloshing.

c) **Synchronization:** Fuel, GPS, and operating-hour data are timestamped and synchronized to ensure consistent data alignment for model evaluation.

d) **Feature Engineering:** Derived features include fuel consumption rate, distance-to-fuel ratio, route segment performance, and driver behavior metrics, which feed into anomaly detection models.

5.2.2 System Methodology

The system was implemented as follows:

- **MCU and Edge Layer:** ESP32/NodeMCU microcontrollers for filtering, timestamping, and initial anomaly scoring.
- **Vehicle Interfaces:** CAN bus or OBD-II for fuel-level readings; GPS module for location and speed tracking.

- **Communication Layer:** LoRa transceivers for real-time cloud updates; GSM backup for critical alerts.
- **Cloud Layer:** Python-based ML models for anomaly detection deployed on cloud servers; REST APIs for data ingestion; PostgreSQL/NoSQL database for structured storage.
- **Visualization Layer:** Web dashboard for real-time monitoring, historical trend analysis, and alert management.

5.2.3 Anomaly Detection and ML Methodology

The anomaly detection workflow is as follows:

a) **Offline Model Training:** Historical datasets are used to train Isolation Forest and Matrix Profile models to detect abnormal fuel usage patterns.

b) **Context-Aware Thresholding:** Thresholds are dynamically adjusted based on vehicle type, route segment, and driver behavior to reduce false positives.

c) **Evaluation Metrics:** Precision, FPR, accuracy, and detection latency are calculated; confusion matrices, ROC, and PR curves visualize performance.

5.2.4 IPO Chart and Design Pipeline

The data design outlines a cohesive pipeline which integrates sensor acquisition, synchronization, cloud transmission, and model-based analysis to enable real-time anomaly detection in fleet fuel systems. Data collection begins with embedded sensors capturing raw signals such as fuel level, fuel flow, GPS coordinates, and ignition events. These signals are preprocessed using filters and signal conditioning circuits to reduce noise and ensure clean input for analysis.

To ensure temporal alignment, all sensor outputs are timestamped and resampled onto a uniform time grid using linear interpolation. This synchronization step is critical for comparing trends across sensor modalities and detecting correlated anomalies. The processed signals are then uploaded to a cloud-based platform for storage and remote access.

For anomaly detection, the synchronized time-series data are used to train and evaluate unsupervised models such as Isolation Forest and Matrix Profile. These models are tested on a dataset composed of real-world and simulated scenarios to reflect both normal and anomalous behavior. Key performance metrics (including MAPE, timestamp drift, detection latency, precision, and false positive rate) are used to assess system effectiveness. This pipeline supports scalable, automated monitoring while laying the groundwork for further model refinement in future stages.

5.2.5 IPO Chart

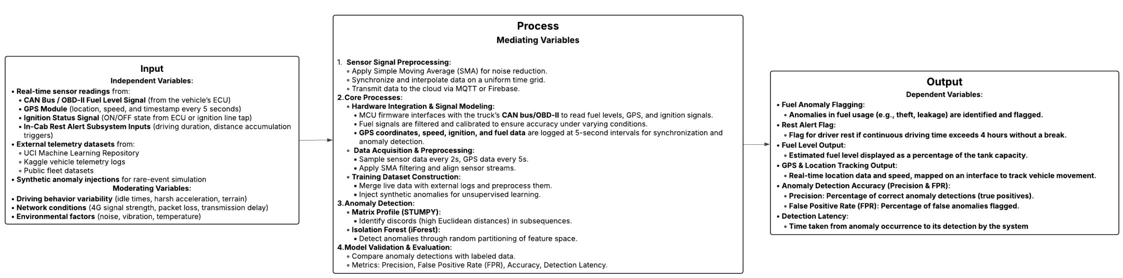

Fig. 5.3 IPO Chart

The IPO (Input-Process-Output) chart provides a structured overview of the IoT-based fleet fuel monitoring and anomaly detection system (Figure 5.3). The input data consists of real-time readings from the truck's CAN bus/OBD-II fuel signal, GPS module capturing coordinates and speed at 5-second intervals, ignition status, and in-cab rest alert subsystem signals. These inputs are supplemented by historical fuel anomaly datasets and synthetic anomalies to simulate rare or edge-case events. Moderating variables such as driving behavior (idle times, acceleration patterns, terrain), network conditions (signal strength, transmission delays), and environmental factors (temperature, vibration) also influence the data.

The process begins with signal preprocessing, where the MCU filters, calibrates, and timestamps fuel, GPS, and ignition data to reduce noise and ensure synchronization. The preprocessed data is then transmitted through LoRa as the primary communication channel, with GSM acting as a backup for essential alerts, and stored in a cloud-based backend. Anomaly detection is carried out in the cloud using machine learning models, including Matrix Profile (STUMPY) for time-series discord identification and Isolation Forest for feature-based outlier detection. Model validation and evaluation are performed using labeled and test datasets, calculating performance metrics such as precision, false positive rate (FPR), detection latency, and overall accuracy.

The system outputs include fuel anomaly flags, such as potential theft or leakage, and driver rest alerts based on cumulative distance or driving duration. Additionally, it provides real-time GPS tracking on a web dashboard, estimated fuel levels as a percentage of tank capacity, and quantitative performance metrics for operational evaluation. This end-to-end flow ensures accurate, real-time monitoring, proactive safety notifications, and actionable insights for fleet management.

5.3 Evaluation

To evaluate these models, the outputs are validated against manually labeled ground truth data. From this, the number of true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN) are counted. Key metrics are computed as follows:

Precision

$$\text{Precision} = \frac{TP}{TP + FP} \tag{5.1}$$

Precision reflects how many detected anomalies are truly anomalous. A high precision means fewer false alarms, which is important for maintaining trust in the alerting system.

False Positive Rate (FPR)

$$\text{FPR} = \frac{FP}{FP + TN} \tag{5.2}$$

FPR quantifies the likelihood of incorrectly classifying normal behavior as anomalous. Lower FPR values are desired to reduce noise in the system's alerts.

Accuracy Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{5.3}$$

Accuracy provides an overall view of model correctness but can be misleading in imbalanced datasets, which is why it's used alongside precision and FPR.

All these metrics use the following classification outcomes:

$$\begin{aligned}
\text{TP (True Positives)} &= \text{Number of correctly identified anomalies,}\\
\text{FP (False Positives)} &= \text{Number of normal events incorrectly flagged as anomalies,}\\
\text{TN (True Negatives)} &= \text{Number of correctly identified normal cases,}\\
\text{FN (False Negatives)} &= \text{Anomalies that were missed by the model.}
\end{aligned}$$

Detection Latency

$$L_{\text{det}} = T_{\text{detected}} - T_{\text{actual}} \tag{5.4}$$

where:

$$\begin{aligned}
T_{\text{detected}} &= \text{the time the anomaly truly occurred,}\\
T_{\text{actual}} &= \text{the time it was identified by the system.}
\end{aligned}$$

The objective is achieved if the unsupervised system attains at least 90% precision (indicating that the majority of flagged events are correct), a false positive rate not exceeding 10%, and average detection latency under 10 seconds. For interpretability, confusion matrices and performance plots such as ROC and PR curves will be used. Detection events will also be overlaid on synchronized GPS location maps and fuel-level graphs for context-aware validation.

5.4 Summary

This chapter presented a comprehensive methodology for an AI-driven fuel monitoring and anomaly detection system. The approach integrates sensor-level calibration, edge preprocessing, IoT-based communication, cloud ML evaluation, and client-side visualization to achieve accurate, real-time, and actionable insights.

REFERENCES

[iec,] Iec 61131-2:2017. https://webstore.iec.ch/en/publication/31007. Accessed: 2025-11-27.

[iee,] Ieee p7003: Algorithmic bias considerations. https://standards.ieee.org/ieee/7003/11357/. Accessed: 2025-11-27.

[fcc,] Ieee xplore full-text pdf. https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10602499. Accessed: 2025-11-27.

[iso, a] Iso 26262-1:2011. https://www.iso.org/standard/43464.html. Accessed: 2025-11-27.

[iso, b] Iso 39001:2012. https://www.iso.org/standard/44958.html. Accessed: 2025-11-27.

[iso, c] Iso 45001:2018. https://www.iso.org/standard/63787.html. Accessed: 2025-11-27.

[iso, d] Iso 7637-2:2011. https://www.iso.org/standard/50925.html. Accessed: 2025-11-27.

[iso, e] Iso/iec 19510:2013. https://www.iso.org/standard/62652.html. Accessed: 2025-11-27.

[iso, f] Iso/iec 27018:2025. https://www.iso.org/standard/27018. Accessed: 2025-11-27.

[iso, g] Iso/iec/ieee 42010:2022. https://www.iso.org/standard/74393.html. Accessed: 2025-11-27.

[rfc,] Rfc 791: Internet protocol. https://datatracker.ietf.org/doc/html/rfc791. Accessed: 2025-11-27.

[iso, 2015] (2015). Iso 14001:2015 environmental management systems. https://www.iso.org/standard/60857.html. ISO.

[aut, 2019] (2019). Automatic fuel monitoring system. *International Journal of Recent Technology and Engineering*, 8(4S2):348–352.

[iso, 2020] (2020). Iso/ts 15143-3:2020: Earth-moving machinery and mobile machines — worksite data exchange — part 3: Telematics data. https://www.iso.org/standard/76394.html. ISO.

[cec, 2022] (2022). Ce marking – obtaining the certificate, eu requirements. https://europa.eu/youreurope/business/product-requirements/labels-markings/ce-marking/index_en.htm. Accessed: 2025-11-27.

[iso, 2022a] (2022a). Iso/iec 27001:2022 information security management systems. https://www.iso.org/standard/27001. ISO.

[iso, 2022b] (2022b). Iso/iec/ieee 29119-1:2022 software and systems engineering — software testing. https://www.iso.org/standard/81291.html. ISO.

[iso, 2023] (2023). Iso 50001 — energy management systems. https://www.iso.org/iso-50001-energy-management.html. ISO.

[iso, 2024] (2024). Iso/pas 8800:2024 road vehicles — safety and artificial intelligence. https://www.iso.org/standard/83303.html. ISO.

[Ahmed et al., 2017] Ahmed, A. A. I., Mohammed, S. A. E., and Satte, M. A. M. H. (2017). Fuel management system. In *2017 International Conference on Communication, Control, Computing and Electronics Engineering (ICCCCEE)*, pages 1–7.

[Alquhali et al., 2019] Alquhali, A. H., Roslee, M., Alias, M. Y., and Mohamed, K. S. (2019). Iot based real-time vehicle tracking system. In *2019 IEEE Conference on Sustainable Utilization and Development in Engineering and Technologies (CSUDET)*, pages 265–270.

[Alsanad et al., 2020] Alsanad, H. R., Ucan, O. N., Ilyas, M., Khan, A. U. R., and Bayat, O. (2020). Real-time fuel truck detection algorithm based on deep convolutional neural network. *IEEE Access*, 8:118808–118817.

[Aquize et al., 2017] Aquize, V. G., Emery, E., and de Lima Neto, F. B. (2017). Self-organizing maps for anomaly detection in fuel consumption. case study: Illegal fuel storage in bolivia. In *2017 IEEE Latin American Conference on Computational Intelligence (LA-CCI)*, pages 1–6.

[Barbado and Corcho, 2022] Barbado, A. and Corcho, (2022). Interpretable machine learning models for predicting and explaining vehicle fuel consumption anomalies. *Engineering Applications of Artificial Intelligence*, 115:105222.

[Beteto, 2022] Beteto, A. e. a. (2022). Anomaly and cyber fraud detection in pipelines and supply chains for liquid fuels. *Environment Systems and Decisions*, 42(2):306–324.

[Crisgar, 2021] Crisgar, P. V. e. a. (2021). Gps-based vehicle tracking and theft detection systems using google cloud iot core firebase. In *2021 International Symposium on Electronics and Smart Devices (ISESD)*, pages 1–6.

[Dhivyasri and Mariappan, 2015] Dhivyasri, G. and Mariappan, R. (2015). Advanced fuel level tracking system using telematic unit. In *2015 IEEE International Conference on Electrical, Computer and Communication Technologies (ICECCT)*.

[Farzana, 2020] Farzana, F. e. a. (2020). A real-time motion based fuel monitoring technique for vehicle tracking systems. In *2020 Emerging Technology in Computing, Communication and Electronics (ETCCE)*, pages 1–6.

[Fortela, 2023] Fortela, D. L. B. e. a. (2023). Unsupervised machine learning to detect impending anomalies in testing of fuel economy and emissions of light-duty vehicles. *Clean Technologies*, 5(1):418–435.

[G, 2022] G, P. e. a. (2022). Real time automatic vehicle monitoring system using iot. In *2022 8th International Conference on Smart Structures and Systems (ICSSS)*, pages 1–6.

[G, 2023] G, Y. e. a. (2023). Fuel monitoring in bike vehicles based on iot. In *2023 Intelligent*

Computing and Control for Engineering and Business Systems (ICCEBS), pages 1–5.

[Garcia Vega, 2022] Garcia Vega, P. (2022). Development of a method for anomaly detection in time series applied to vehicle monitoring. Master's thesis, Polytechnique Montréal.

[Husak,] Husak, V. e. a. Intelligent real-time vehicle tracking information system. n.d.

[IEEE Standards Association,] IEEE Standards Association. Ieee standards association. https://standards.ieee.org/ieee/1451.0/11001/. Accessed: 2025-11-27.

[Jose, 2015] Jose, D. e. a. (2015). Intelligent vehicle monitoring using global positioning system and cloud computing. *Procedia Computer Science*, 50:440–446.

[Joshi, 2024] Joshi, P. e. a. (2024). Intelligent iot-enabled real-time monitoring system for logistics management. In *2024 International Conference on Computing, Sciences and Communications (ICCSC)*, pages 1–5.

[Komal Shoukat Ali, 2018] Komal Shoukat Ali, K. e. a. (2018). Automatic fuel tank monitoring, tracking theft detection system. In *MATEC Web of Conferences*, volume 184.

[Krishnasamy, 2019] Krishnasamy, R. (2019). Automatic fuel monitoring system. *International Journal of Recent Technology and Engineering (IJRTE)*, 8(4S2):348–352. Accessed: 2025-11-18.

[M, 2021] M, S. e. a. (2021). Digital fuel monitoring system for automobiles. In *2021 Second International Conference on Electronics and Sustainable Communication Systems (ICESC)*, pages 6–10.

[Mathi, 2024] Mathi, S. C. e. a. (2024). Development of an intelligent fuel monitoring and theft detection system equipped with gps & gsm integration, state of charge monitoring and realtime alarm generation. In *2024 International Conference on Innovative Computing and Communication Technologies (ICICT)*, pages 2095–2102.

[Maurya,] Maurya, K. e. a. Real time vehicle tracking system using gsm and gps technology-an anti-theft tracking system. n.d.

[Mukesh, 2010] Mukesh, P. R. (2010). Real time web based vehicle tracking using gps.

[Mumcuoglu, 2023] Mumcuoglu, M. E. e. a. (2023). Fuel consumption classification for heavy-duty vehicles: A novel approach to identifying driver behavior and system anomalies. In *2023 AEIT International Conference on Electrical and Electronic Technologies for Automotive (AEIT AUTOMOTIVE)*, pages 1–6.

[Nada, 2017] Nada, Y. A. (2017). Design and implementation for trucks tracking system using gps based on semantic web. *International Journal of Trend in Scientific Research and Development*, 1(4).

[NASA EarthData, 2025] NASA EarthData (2025). Global navigation satellite system. https://www.earthdata.nasa.gov/data/space-geodesy-techniques/gnss. Accessed: 2025-11-27.

[Obikoya, 2014] Obikoya, G. D. (2014). Design, construction, and implementation of a remote fuel-level monitoring system. *EURASIP Journal on Wireless Communications and Networking*, 2014(1):76.

[Patil and Patil, 2024] Patil, A. and Patil, S. (2024). Iot enabled vehicle monitoring and theft prevention system (i-vmtps). In *2024 2nd International Conference on Advancements and Key Challenges in Green Energy and Computing (AKGEC)*, pages 1–7.

[Prabhu et al., 2022] Prabhu, S. B., Deepa, K., and Nithya, M. (2022). Iot enabled fuel level monitoring and automatic fuel theft detection system. In *2022 13th International Conference on Computing Communication and Networking Technologies (ICCCNT)*.

[Prasanna et al., 2023] Prasanna, J. L., Santhosh, C., Kumar, M. R., and Gayathri, K. (2023). Iot based automatic fuel level monitoring system for automobiles. In *IET Conference Proceedings*, volume 2023, pages 56–61.

[RS et al., 2024] RS, S., Narayanan CS, S., G, R., S, P., Alagarsamy, M., and William, P. (2024). Fuel quantity and quality detection in automobiles using iot. In *2024 4th International Conference on Innovative Practices in Technology and Management (ICIPTM)*, pages 1–6.

[S et al., 2024] S, S., Sowmya, J. M., Siva, M. V., Rishitha, N., Sagar, M. K. G. S., and Ganesh, M. (2024). Iot enabled smart fuel station management system. In *2024 2nd International Conference on Networking and Communications (ICNWC)*, pages 1–5.

[Samainstage, 2025] Samainstage (2025). The evolution of wi-fi technology and standards. https://standards.ieee.org/beyond-standards/the-evolution-of-wi-fi-technology-and-standards/. Accessed: 2025-11-27.

[Semenov et al., 2024] Semenov, I., Świderski, A., Borucka, A., and Guzanek, P. (2024). Concept of early prediction and identification of truck vehicle failures supported by in-vehicle telematics platform based on abnormality detection algorithm. *Applied Sciences (Switzerland)*, 14(16).

[Tatababu et al., 2024] Tatababu, D., Ranga, M. R., Satyannarayana, D., Jyothi, Y., and Devi, A. R. (2024). Smart fuel monitoring system for diesel generator. In *2024 International Conference on Computational Intelligence for Green and Sustainable Technologies (ICCIGST)*, pages 1–6.

[Thajudheen et al., 2023] Thajudheen, S., Shanmugarathinum, G., and Jesudoss, A. G. (2023). Vehicular data retrieval system. In *2023 International Conference on New Frontiers in Communication, Automation, Management and Security (ICCAMS)*.

[Turan et al., 2024] Turan, B. B., Gene, E., Akcig, I. N., Goztepe, N., Mumeuoglu, M. E., and Unel, M. (2024). Detecting high fuel consumption in hdvs with ensemble of anomaly detection models. In *2024 IEEE 22nd International Conference on Industrial Informatics (INDIN)*, pages 1–6.

[Uddin et al., 2017] Uddin, M. S., Ahmed, M. M., Alam, J. B., and Islam, M. (2017). Smart

De La Salle University

anti-theft vehicle tracking system for bangladesh based on internet of things. In *2017 4th International Conference on Advances in Electrical Engineering (ICAEE)*, pages 624–628.

[Usama et al., 2019] Usama, M., Qadir, Raza, A., Arif, H., Yau, A., Elkhatib, Hussain, and Al-Fuqaha, A. (2019). Unsupervised machine learning for networking: Techniques, applications and research challenges. *IEEE Access*, 7:65579–65615.

[Vigneshwaran et al., 2020] Vigneshwaran, S., Nithya, B., Raghul, K., Nivas, B., and Kishore, V. M. (2020). Design of bus tracking and fuel monitoring system. In *2020 6th International Conference on Advanced Computing and Communication Systems (ICACCS)*, pages 348–351.

[Weiwei et al., 2024] Weiwei, Q., Longbin, L., Xingyu, W., Yulong, L., and Weiqian, D. (2024). Research on diesel consumption prediction and early warning analysis of dump trucks in smart mines. In *2024 IEEE 22nd International Conference on Industrial Informatics (INDIN)*, pages 1–6.